



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-2026

Semester: I

Examination: ISE I

Class: T.Y. (CSE)

Branch: CSE

Date: 20/08/2025

Subject: Design and Analysis of Algorithms

Code: PCC-CS305

Max. Marks: 20

Time: 15:00 To 15:45

Solve All Questions

Q.1.	(a)	Write and explain the Master Theorem in detail using suitable examples for each case. (04)	(10)												
	(b)	Solve the following recurrences using Master Theorem: (06) i) $T(n) = 3 T\left(\frac{n}{2}\right) + n^2$ ii) $T(n) = 2^n T\left(\frac{n}{2}\right) + n^n$ iii) $T(n) = 2 T\left(\frac{n}{2}\right) + 10n$													
Q.2.	Write and explain the deterministic Quick Sort algorithm. Also, prove its worst-case time complexity. Trace Quick Sort on A = [50, 30, 70, 90, 10, 40] using the first element as the pivot.		(5)												
Q.3.	What is the notion of Optimization Problem and Greedy Paradigm? Write and explain the simple knapsack problem and the greedy algorithm for it. Solve the following: Assume a knapsack with a maximum weight capacity of 10 kg. You have five items to choose from: <table><tr><th>Item</th><th>Weight (kg)</th></tr><tr><td>A</td><td>6</td></tr><tr><td>B</td><td>5</td></tr><tr><td>C</td><td>4</td></tr><tr><td>D</td><td>2</td></tr><tr><td>E</td><td>1</td></tr></table> Which items should you pack to carry the maximum number of objects without exceeding the given capacity limit.		Item	Weight (kg)	A	6	B	5	C	4	D	2	E	1	(5)
Item	Weight (kg)														
A	6														
B	5														
C	4														
D	2														
E	1														

Course Outcomes (COs):

- CO1: Analyze the efficiency of algorithms by determining their time and space complexities.
- CO2: Apply Divide and Conquer, Greedy, Dynamic Programming, and Backtracking techniques to solve complex problems effectively.
- CO3: Implement algorithms to solve optimization problems like Knapsack, Job Sequencing, Matrix Chain Multiplication, and Traveling Salesman Problem.
- CO4: Design and analyze graph algorithms for shortest path, spanning tree, connectivity, and Eulerian paths.
- CO5: Differentiate between P, NP, NP-Hard, and NP-Complete problems, and demonstrate understanding through appropriate examples.
- CO6: Apply appropriate algorithmic techniques to design optimal or near-optimal solutions for computationally challenging problems.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create

Question No	CO No	BT No
1	1	1, 2, 3
2	2	1, 2, 3
3	2, 3	1, 2, 3



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year : 2025-26 (Semester I) Class : T. Y. B. Tech (CSE)
In Semester Evaluation – 1 Examination
Subject : Operating Systems Code : PCC-CS304
Date: 18/Aug/2025 Time: 15:00 to 15:45 Max. Marks :20

- Note:**
1. Please read the questions **carefully**.
 2. Marks are shown at the end of each question on the right hand side .
 3. Marks will be deducted if answers are not written in **Neat handwriting** .
 - 4). Do not write **essay**.

- 1). Draw the functional diagram of **Intel i7 CPU**. Show the various cache memories available in **i7 CPU**. Explain how CPU makes use of Cache Memories.

What is the meaning of Cache-line ? What is the cache line size in i7?

Write down the approximate **access times** of the cache memories and the main memory. Write the formula to calculate average access time in a Memory hierarchy. [7]

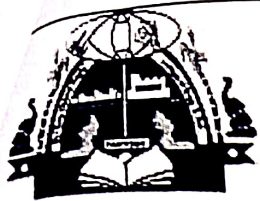
- 2). Write a c language program to create **20 pthreads**. Each thread executes a function named **myfn()**. This function increments the value of the global variable **MYVAR 100,000 times**. After all the threads complete their work , the main thread displays the contents of MYVAR.

Explain the reason for actual output.

What changes should be made to get correct output? Show the modified code. [7]

- 3). Write and explain a simple c code to create a process.
What is TLS in a thread?

[5]
[1]



SHRI GURU GOBIND SINGHJI INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED

Department Of Computer Science & Engineering

T. Y. B. Tech(CSE)

In-Semester Evaluation-I (ISE I - Year 2025-2026)

Date: 19th August 2025

Time: -03:00-03:45

Subject: - Formal Language and Automata Theory (PCC-CS303)

Marks-20

Q.1 Solve the following questions sequentially and assume data if necessary.

Questions.....	Marks	CO(BT)																	
Q.1-a) If $A=\{1,2,3\}$ and $B=\{2,3,4\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.	2	CO1(BT1)																	
b) Prove by contradiction: $\sqrt{2}$ is irrational.	2	CO1(BT1)																	
Q.2-a) Differentiate between DFA and NFA.	2	CO2(BT2)																	
Consider the following Deterministic Finite Automaton, M. Let $L(M)$ denote the set of eight-bit strings whose second, third, sixth, and seventh bits are 1. How many strings in that are accepted by M? OR Construct a finite automata to accept binary strings that represent numbers divisible by 4. Explain by taking one example.	3	CO1(BT4)																	
b) For the finite state machine M given in the following table, test whether the strings "101101" and "11111" are accepted by M (q_0 is the initial and final state). <table border="1"> <thead> <tr> <th rowspan="2">State</th><th colspan="2">Input</th></tr> <tr> <th>0</th><th>1</th></tr> </thead> <tbody> <tr> <td>q_0</td><td>q_0</td><td>q_1</td></tr> <tr> <td>q_1</td><td>q_3</td><td>q_0</td></tr> <tr> <td>q_2</td><td>q_0</td><td>q_3</td></tr> <tr> <td>q_3</td><td>q_1</td><td>q_2</td></tr> </tbody> </table>	State	Input		0	1	q_0	q_0	q_1	q_1	q_3	q_0	q_2	q_0	q_3	q_3	q_1	q_2	5	CO2(BT3)
State		Input																	
	0	1																	
q_0	q_0	q_1																	
q_1	q_3	q_0																	
q_2	q_0	q_3																	
q_3	q_1	q_2																	
c) Draw an NFA and a DFA that accepts all strings over $\{0,1\}$ ending in "0001" and show the instantaneous description (e.g, $\delta(q_0,1) = \text{next state}$) for the string "1010001" by using the resultant NFA and DFA.	6	CO2(BT6)																	

***** All the Best *****

CO's	Statement
CO1	Understand the fundamental concepts of automata and formal languages.
CO2	Analyze and construct finite automata (FA) and regular expressions (RE).
CO3	Demonstrate knowledge of context-free grammars (CFGs) and context-free languages (CFLs).
CO4	Examine pushdown automata (PDA) and their relationship with context-free grammars.
CO5	Understand how Turing machines (TM) and computational theory work
CO6	Explore the applications and limitations of formal language theory.
BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create	



SGGS Institute of Engineering and Technology, Vishnupuri,
Nanded

Academic Year: 2025-2026

In-Semester Evaluation-I

Class: Third Year (All Branch)

Date: 25/08/25
Time: 15:00-15:45 Hrs

Subject: Introduction to Cryptography
Code: MDM-IS-CS301

Max Marks: 20

Q.N.	Questions	Marks	CO	BT																																																																																					
1.	Define cryptography and give its significance in today's world? How do symmetric and asymmetric cryptography differ?	[6]	CO2	BT1																																																																																					
2.	a. Explain the fundamental concept behind a Feistel network, including how it divides plaintext and processes it through multiple rounds. OR b. Explain 1. Differential Cryptanalysis 2. Meet in the middle attack	[8] [8]	CO2 CO2	BT1 BT2																																																																																					
3.	1) Identify valid or invalid congruences a) $7 \equiv -11 \pmod{6}$ b) $97 \equiv 19 \pmod{12}$ 2) Find the multiplicative inverse of 23 in Z_{100} 3) Using below given S-box 1 of DES , find the output for the input 110011. Show the row, column selection, and final binary result. <table border="1"><tr><td></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>0</td><td>14</td><td>04</td><td>13</td><td>01</td><td>02</td><td>15</td><td>11</td><td>08</td><td>03</td><td>10</td><td>06</td><td>12</td><td>05</td><td>09</td><td>00</td><td>07</td></tr><tr><td>1</td><td>00</td><td>15</td><td>07</td><td>04</td><td>14</td><td>02</td><td>13</td><td>10</td><td>03</td><td>06</td><td>12</td><td>11</td><td>09</td><td>05</td><td>03</td><td>08</td></tr><tr><td>2</td><td>04</td><td>01</td><td>14</td><td>08</td><td>13</td><td>06</td><td>02</td><td>11</td><td>15</td><td>12</td><td>09</td><td>07</td><td>03</td><td>10</td><td>05</td><td>00</td></tr><tr><td>3</td><td>15</td><td>12</td><td>08</td><td>02</td><td>04</td><td>09</td><td>01</td><td>07</td><td>05</td><td>11</td><td>03</td><td>14</td><td>10</td><td>00</td><td>06</td><td>13</td></tr></table>		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07	1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08	2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00	3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13	[6]	CO1 CO1 CO1	BT5 BT5 BT3
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